

Attachment A

**INDEFINITE DELIVERY - INDEFINITE QUANTITY
CONTRACT**

WOOD CROSSARMS AND BRACES SPECIFICATIONS

February 2017



SUPPLIES OR SERVICES AND PRICES/COSTS

The Western Area Power Administration intends to award an Indefinite Delivery/Indefinite Quantity (IDIQ) contract as a result of this solicitation, for wood crossarms and braces for the Upper Great Plains, Desert Southwest, and Rocky Mountain Regions.

The award period shall be for a one year base period, with two one-year option periods.

The minimum guaranteed amount for the duration of the contract, including all options, shall be \$25,000.00. The maximum dollar amount, provided all options are exercised, is estimated to be \$9,000,000.00.

All supplies delivered under delivery orders issued against this contract shall be delivered within 6-8 weeks after receipt of the delivery order.

The contractor shall furnish and deliver to Western Area Power Administration (also referred to as Western), the following described items, all complete in accordance with attached specifications:

(continued on next page)

Specification for Solid Sawn and Laminate Wood Crossarms

General.

Furnish and deliver crossarms or crossarm sets as stated under this technical specification. Materials shall be of the type and grade specified and shall conform to industry standards and practices. All materials shall be new and free from defects.

Special Requirements.

Refer to Attachment E "Technical Specification for Western Area Power Administration operation and maintenance purchase of crossarms and braces" dated February 2017 for solid sawn and laminate requirements.

Knot restrictions apply as outlined in the notes of Drawing No. 41-6019c and 41-6020b as applicable. Crossarm dimensions and drill locations are specified in accordance with Drawing Nos. 41-6113 and 41-6114a as applicable for use with adjustable or fixed spacer fittings are specified.

Should double crossarm units or "sets" be specified, the individual crossarms shall be assembled into "double crossarm" units as shown in Drawing Nos. 41-6113 and 41-6114 as applicable. Each double crossarm unit will consist of two crossarms and three adjustable spacer fittings complete with bolts, nuts and washers. Note that threads on vertically oriented bolts will be up and on horizontally oriented bolts will be to the outside of the crossarm. Note that the crossarm sets come complete with the twisted clip (equal to Hughes Bros. catalog # AS2276-____ or equal) and bent bolt on the outside phase fittings and a bent stud in the center fitting.

Hardware.

- (1) Galvanizing - All ferrous metal shall be galvanized in accordance with ASTM Designation A153. All metal shall be free from weld spatter and flux, burrs, sharp edges, and dross and shall be smooth so that inter-connecting parts will fit properly. All threaded parts shall be galvanized after being threaded, and excess zinc shall be removed from the threads. All nuts and locknuts shall be re-tapped after being galvanized and shall be capable of being threaded the entire length of the threads without the use of a wrench.
- (2) Bolts, nuts, locknuts, and lag screws - All bolts, nuts, locknuts, and lag screws shall be made of steel which will meet the requirements of new-rolled, open-hearth steel conforming to Federal Specifications FF-B-571, Class C steel. Rolled threads of the American coarse-thread series shall be furnished on all bolts, nuts, and locknuts.

Storage, Handling, and Transport:

1. Individual crossarms may be bundled and banded together in such a manner that when placed in outdoor storage they will not come in contact with a hard surfaced storage area, and can be stacked two bundles high with no damage to the crossarm surfaces or crushing of spacers between crossarms.
2. Individual crossarms shall have a minimum 1-inch thick spacer between each layer of the bundle for air circulation while in storage.
3. Crossarm sets may be bundled and banded together in such a manner that when placed in outdoor storage they will not come in contact with a hard surfaced storage area, and can be stacked two bundles high with no damage to the crossarm surfaces,

Inspection.

All work performed may be subject to inspection, and no crossarms shall be shipped until all required inspections have been made. The contractor shall furnish reasonable facilities, including skids, blocks, sufficient space and personnel to provide for laying out in an orderly manner and handling and turning a reasonable number of crossarms at a time for inspection. Inspection will be made before and after treatment unless one or both inspections are waived. The Government Inspector shall have the right to take samples of the preservative solution or ingredients thereof whenever he may elect to do so. Such samples shall be analyzed whenever the Government may desire to have the analysis made. Acceptance of the material or the waiving the inspection thereof shall in no way relieve the contractor of the responsibility for furnishing materials meeting the requirements of the specifications.

SPECIFICATION FOR SOLID SAWN AND LAMINATE X-BRACES, KNEE BRACES, & VEE BRACES

General.

Laminate material and treatment specifications within section 'Specification for Laminated Crossarms' shall apply to all X-braces, Knee Braces, and Vee Braces with crosssections of 5-1/8 x 6-0 inches and greater.

Special Requirements,

Individual X-braces shall include a single brace complete with end fittings. X-brace assemblies shall be of the two-piece type complete with assembled wood sections, end fittings, bolts, washers, and center clamp. The bolts or rods for the center clamp shall be constructed so that space will be left to permit free movement of the wood members after the clamp is tightened. The wood members of the x-braces, knee braces, and vee braces shall have a cross section at minimum dimensions defined below and per reference drawings.

All X-braces, knee braces, crossarms, and vee braces shall be stamped in metal and nailed to wood product or painted to define structure pole spacing center to center dimension and manufacturers' product number. Manufacturer brand or painted name is not permitted on crossarms or braces legible past distance to 20 feet.

X-braces having cross sections of 3-3/8 x 4-3/8 inches and 3-3/8 x 5-3/8 inches shall be complete with four 7/8-inch machine bolts suitable for attaching to the poles, 7/8-inch square Type "MF" locknuts, and 3-1/2 x 3-1/2 x 3/8-inch square curved washers for 7/8-inch bolts. X-braces having cross sections of 5-1/8 x 6-0 inches shall be complete with mounting bolts, 4" x 4" x 3/8" square curved washer, tees, pin bolts, and center clamps.

69-kV X-braces shall accommodate 10'-0" pole spacing (Equal to Hughes Bros. Catalog #1042-10-0-CPT or equal). 115-kV X-braces shall accommodate 12'-0" pole spacing (Equal to Hughes Bros. Catalog #1042-12-0-CPT or equal), 161-kV X-braces shall accommodate 17'-0" pole spacing (Equal to Hughes Bros. Catalog #1042-17-0-CPT or equal).

230-kV X-Braces shall accommodate the following pole spacings:

- 22'-0" pole spacing (Equal to Hughes Bros. Catalog # 2121-22-0-CPT or equal).
- 23'-0" pole spacing. (Equal to Hughes Bros. Catalog # 2121-23 0-CPT or equal).
- 24'-0" pole spacing. (Equal to Hughes Bros. Catalog # 2121-24-0-CPT or equal).

The knee and vee braces shall be complete with end fittings. 115-kV knee and vee braces shall have 5'-8" spacing and 5'-8" drop (Equal to Hughes Bros. Catalog # 2025BDV-110, 2025BDK-110 or equal). 161-kV knee and vee braces shall have 8'-2" spacing and 6'-0" drop (Equal to Hughes Bros. Catalog # 2025BDV-122, 2025BDK-122 or equal). 230-kV vee braces shall have 10'-8" spacing and 8'-6" drop (Equal to Hughes Bros. Catalog # 2579BDV-162 or equal). 230-kV vee braces for 3AC-1 structure (dwg 41-6131-1 and 41-6131-2) shall have a 24'-9" C-C length with pin type connection (Equal to Hughes Bros. Catalog # 2121-19-0PC or 2121-19-0PC-CPT (complete with tees, mounting bolts, washers) as specified).

Hardware.

(1) Galvanizing – All ferrous metal shall be galvanized in accordance With ASTM Designation: A153, All metal shall be free from weld spatter and flux, burrs, sharp edges, and dross and shall be smooth so that inter-connecting parts will fit properly. All threaded parts shall be galvanized after being threaded, and excess zinc shall be removed from the threads. All nuts and locknuts shall be re-tapped after being galvanized and shall be capable of being threaded the entire length of the threads without the use of a wrench.

(2) Bolts, nuts, locknuts, and lag screws - All bolts, nuts, locknuts, and lag screws shall be made of steel which will meet the requirements of new-rolled, open-hearth steel conforming to Federal Specifications FF-B-571, Class C steel. Rolled threads of the American coarse thread series shall be furnished on all bolts, nuts, and locknuts.

Inspection.

All metals furnished and all work performed shall be subject to inspection, and no braces shall be shipped until all required inspections have been made. The contractor shall furnish reasonable facilities, including skids, blocks, sufficient space and personnel to provide for laying out in an orderly manner and handling and turning a reasonable number of braces at a time for inspection.

Inspection will be made before and after treatment unless one or both inspections are waived. The Government Inspector shall have the right to take samples of the preservative solution or ingredients thereof whenever he may elect to do so. Such samples shall be analyzed whenever the Government may desire to have the analysis made. Acceptance of the materials or the waiving of the inspection thereof shall in no way relieve the contractor of the responsibility for furnishing materials meeting the requirements of the specifications.

Type HS-1 115-kV Suspension Structure				
Drawing 41-6122B				
Bundled Framing Kit Required Material List				
Part No.	Quantity	Description	Hughes Bros. Product Number (or equal)	Other
2, 3, 10, 11	1	Crossarm Assembly: (2) glue laminated wood crossarms, (3) adjustable spacer fittings complete with bolts, nuts, and washers, (2) 7/8" bent bolts, (1) 7/8" bent stud, (2) twisted clips.	See specification	Crossarm is laminate, Vertical through bolt length to be specified.
4	2	Double Arming Bolt, 7/8" x 24" length, with 4 nuts	TR-8LengthF	24" Bolt length
5	1	X-Brace Set, complete with fittings	1042-12-0-CPT	Bolt length at pole to be specified
6	4	Flat Gain Plate for 9" Arm	1004.895	
7	4	Ribbed Tie Plate for 9" Arm	1005.895	
9	2	Clevis-Clevis Fiberglass Guy Strain, 25,000 lb. minimum ultimate strength, complete with bolt, nut, key.	CF695-36-BNK	
10	2	Twisted Clip (at pole)	AS2276-Angle	Angle to be specified
12	2	7/8" x 12", Square Head Machine Bolt with nut	BBLength-6	For knee brace to pole. 12" Bolt length
13	4	4" x 4" x 1/4" Curved washer, 15/16" hole	SCW4-80 or CW80	For (2) knee brace to pole, (2) steel angle to pole.
14	2	Turnbuckle with jaw & rod end eye	AS2545-B	
N/A	3	Ball Y-Clevis 30k	Hubbell YBC-30	
Overhead Ground Wire and Grounding Material				
22	2	7/8" x 12" Square Head Machine Bolt with nut	B812-6	For steel angle to pole
28	4	Locknut for 7/8" Bolt	MF80	For (2) steel angle to pole, (2) vee brace to pole
23,24,32	1	Steel Angle, 4" x 3" 1/2" x 5/16" x 14'-6" long, 15/16" holes at pole. Complete w/(2) U-Bolts (3/4" x 3", w hex nuts, locknuts) & (2) Chain Links (5/8")	C4373.36A	

Type HS-1 230-kV Suspension Structure				
Drawing 41-6128D				
Bundled Framing Kit Required Material List				
Part No.	QTY	Description	Hughes Bros. Product Number (or equal)	Other
2, 3, 10, 11	1	Crossarm Assembly: (2) 3-1/8" x 9" x 45' glue laminated wood crossarms, (3) adjustable spacer fittings complete with bolts, nuts, and washers, (2) 7/8" bent bolts, (1) 7/8" bent stud, (2) twisted clips.	C4342.6BL	
4	2	Double Arming Bolt, 7/8" x required length, with 4 nuts	TR-8LengthF	Bolt length to be specified
5	1	X-Brace Set, complete with fittings	2121-22-0-CPT	Bolt length at pole to be specified
6	4	Flat Gain Plate for 9" Arm	1004.895	
7	4	Ribbed Tie Plate for 9" Arm	1005.895	
9	2	Clevis-Clevis Fiberglass Guy Strain, 30,000 lb. minimum ultimate strength, complete with bolt, nut, key.	CF695-115-BNK	
10	2	Twisted Clip (at pole)	AS2276-Angle	Angle to be specified
11	2	Wood Vee Brace. 3-5/8" x 7-1/2" complete w/fittings. 10'-8" spacing, 8'-6" drop	2579BDV-162	
12	2	Turnbuckle with jaw & rod end eye	AS2545-B	
14	2	7/8" x Required Length, Square Head Machine Bolt with nut	B8Length-6	For vee brace, Bolt length to be specified
Overhead Ground Wire and Grounding Material				
21	2	Curved Washer, 4" x 4" x 1/4", 15/16" hole	SCW4-80 or CW80	
22	2	7/8" x 12" Square Head Machine Bolt with nut	B812-6	For steel angle to pole
28	8	Locknut for 7/8" Bolt	MF80	For (2) steel angle to pole, (4) bonding clip at crossarm & X-brace, (2) vee brace to pole
23, 24, 32	1	Steel Angle, 4" x 3" 1/2" x 3/8" x 25'-6" long, 15/16" holes at pole. Complete w/U-Bolts (3/4" x 3", w hex nuts, locknuts) & Chain Links (5/8")	C4373.36A	
34	4	Bonding Clip, for 7/8" bolt diameter.	2727-8	For bonding at crossarm & X-brace

Specification for Bundled Framing Kits

General

Furnish and deliver bundled structure framing kits as stated in this technical specification.

Material and Hardware

All material and hardware shall conform to specifications and standards set forth within "Specification for Laminated Crossarms" and "Specification for Solid Sawn and Laminate X-Braces, Knee Braces, & Vee Braces".

Fiberglass guy strains shall be Fiberglass Reinforced Plastic (FRP). The FRP shall be a polyester/vinylester matrix containing ultraviolet protective inhibitors. All exposed fiberglass rod shall be coated with 2-3 mils of acrylic urethane enamel paint baked to hardness. Each guy strain shall be factory tested to 100% of its rating.

Special Requirements

Bundled framing kits shall be provided for type HSB-1 230kV Suspension Structure (drawing 41-6128D) and type HS-1115kV Suspension Structure (drawing 41-6122B) as applicable. The bundled kit for each structure shall include all materials as listed.

Packaging

Each bundle shall contain all material required for one structure, bundled and banded together. All packaging material, including banding, shall be suitable for long term (Minimum of 10 years) outdoor storage.